

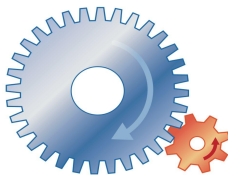
The background features abstract geometric shapes in shades of blue and red. In the top left, there's a dark blue shape with diagonal lines. In the top right, a red shape with diagonal lines. The bottom left shows a light blue grid pattern. The bottom center has a blue shape with vertical lines. The bottom right has a blue shape with horizontal lines.

DevSecOps

By : Syed Azhar

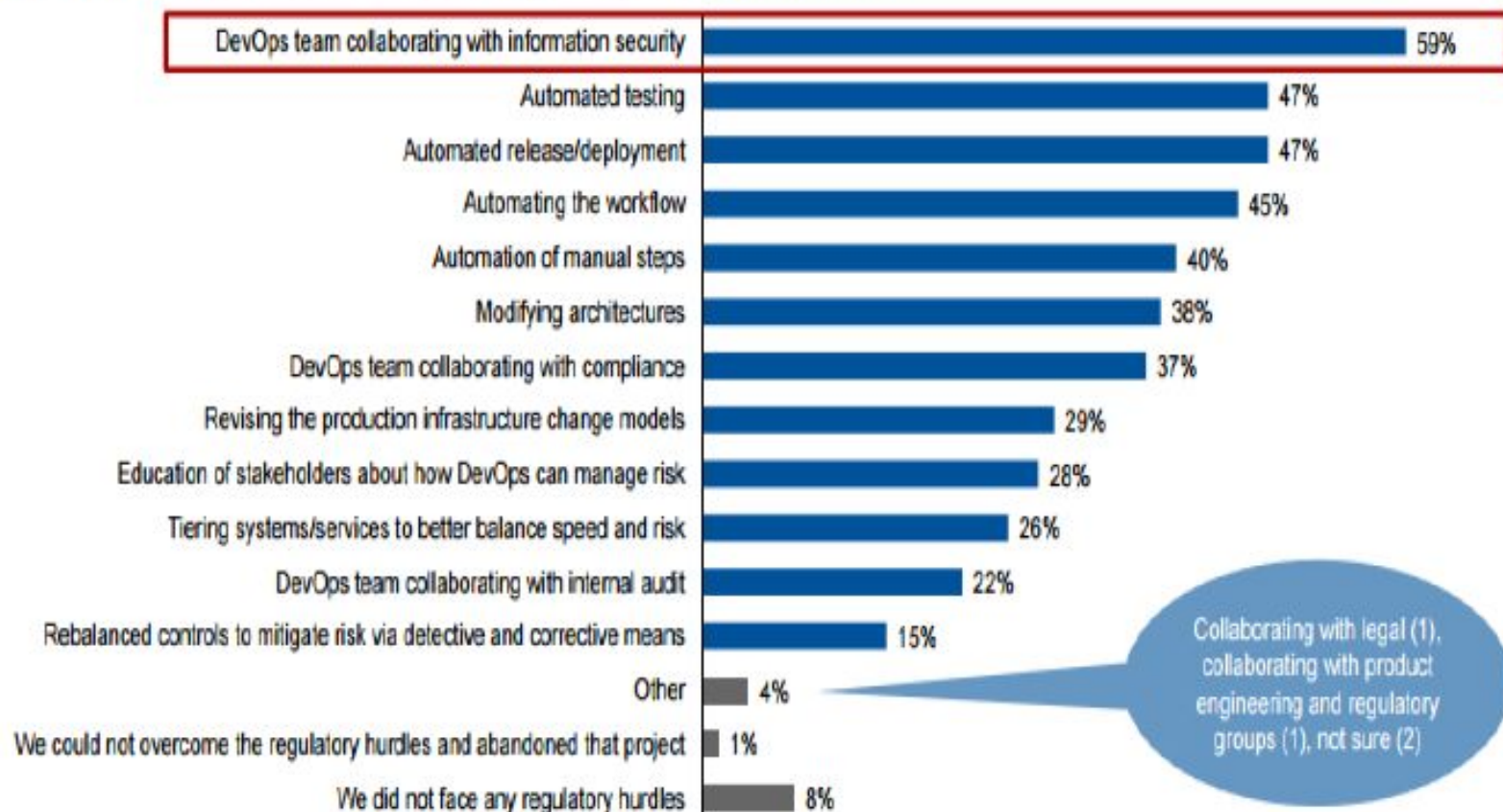
DevSecOps

Everyone is responsible for security



DevSecOps & Compliance

Strategies to overcome hurdles related to using DevOps in regulated situations:



Better, faster, cheaper software with DevOps

COMPANY	DEPLOY FREQUENCY	DEPLOY LEAD TIME	RELIABILITY	CUSTOMER RESPONSIVENESS
AMAZON	23000 / day	Minutes	High	High
GOOGLE	5500 / day	Minutes	High	High
NETFLIX	500 /day	Minutes	High	High
FACEBOOK	1 / day	Minutes	High	High
TWITTER	3 / week	Minutes	High	High
TYPICAL ENTERPRISE	Once in every 9 Months	Months or quarters	Low / Medium	Low / Medium

but is it secure?

Meet DevSecOps

DevSecOps enables organisations to deliver the secure software at DevOps speed.

DevSecOps seeks to achieve greater efficiency and productivity by incorporating security principles within DevOps process.

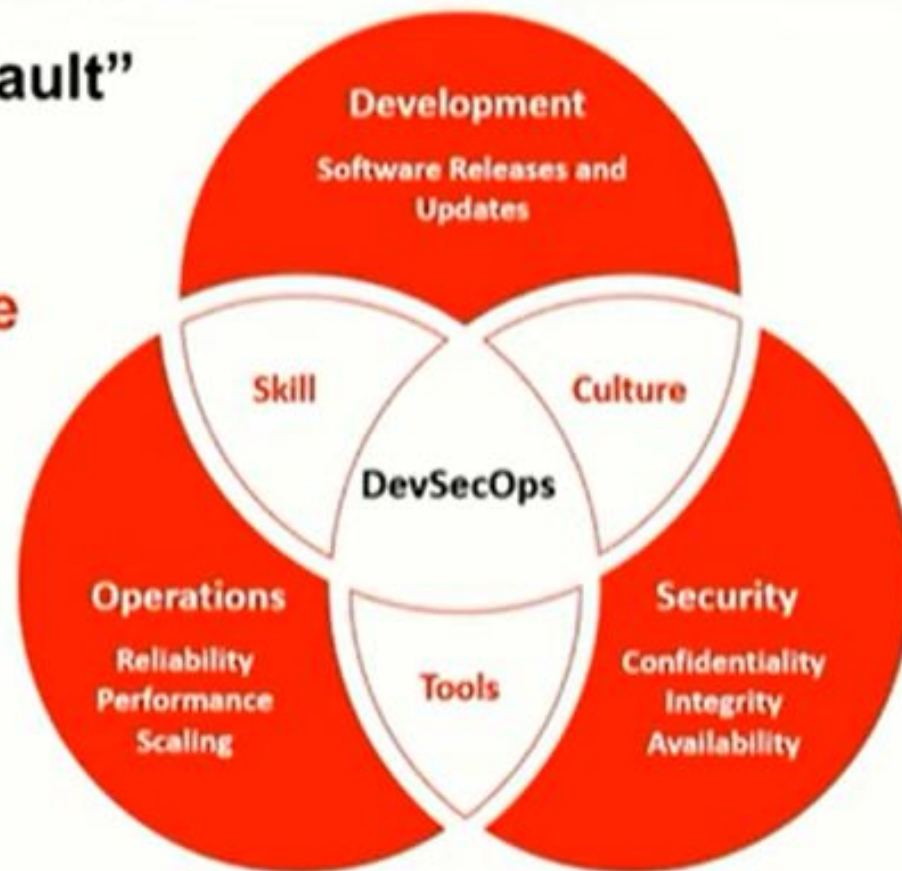
DevOps + Security : **DevSecOps**



What is Devops?

Effort to strive for “Secure by Default”

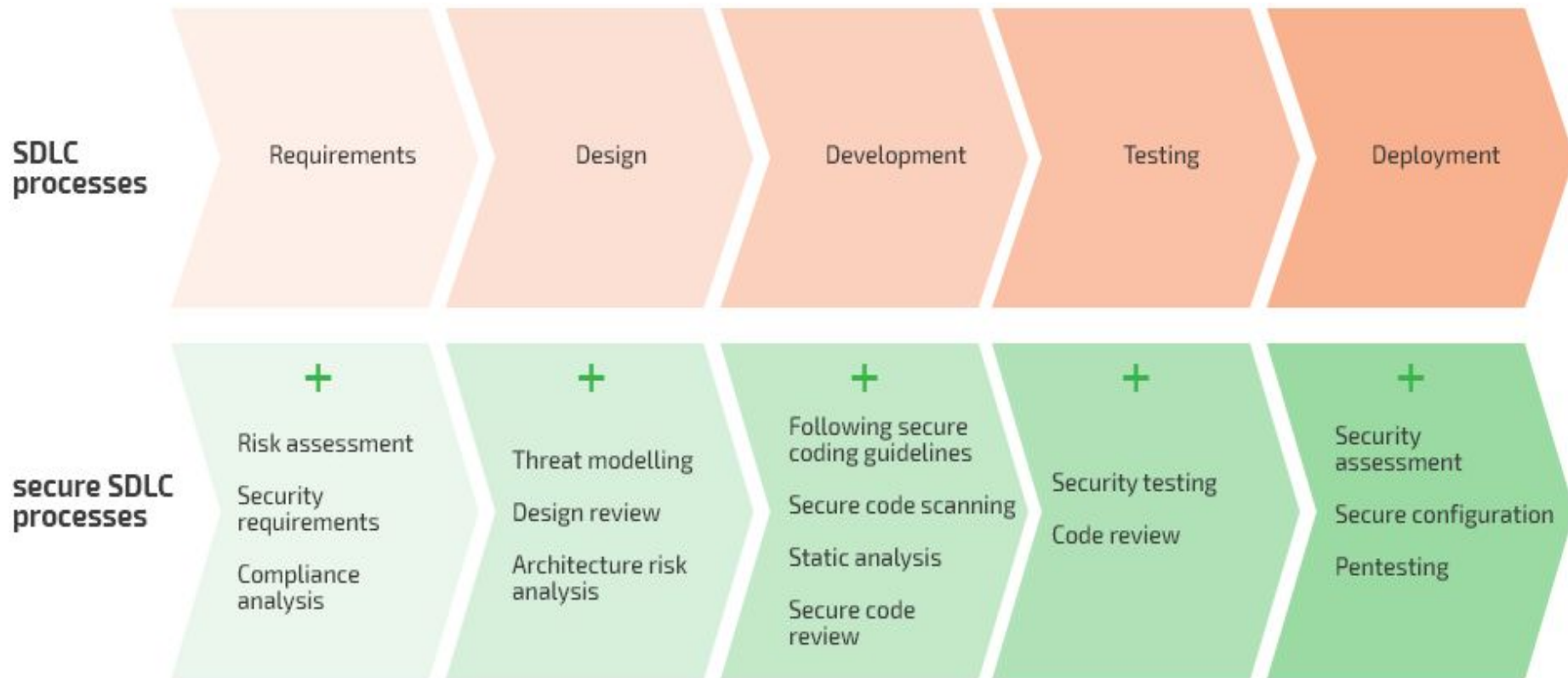
- Integrate Security via **tools**
- Create Security as Code **culture**
- Promote cross **skilling**



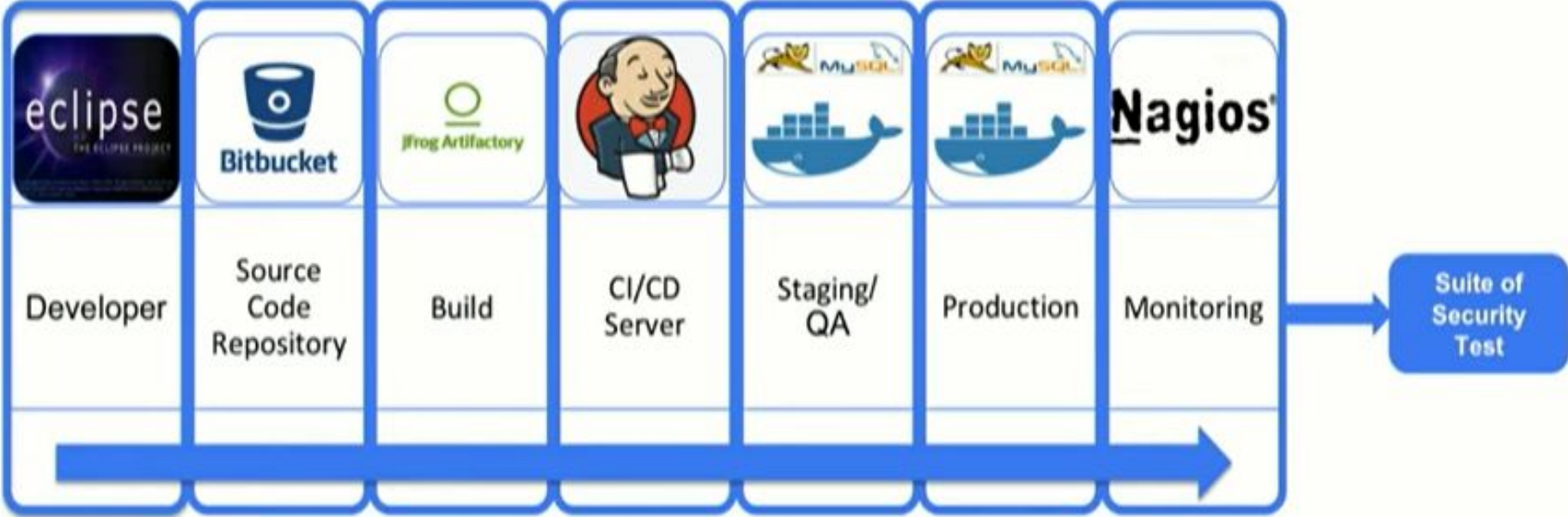
Why we need DevSecOps?

- **DevOps moves at rapid pace, traditional security just can't keep up**
- **DevSecOps makes it easier to manage rapid pace of development & large scale secure deployments**
- **DevSecOps allows for much smoother scaling of process**
- **Security as part of process is the only way to ensure safety**

SDLC

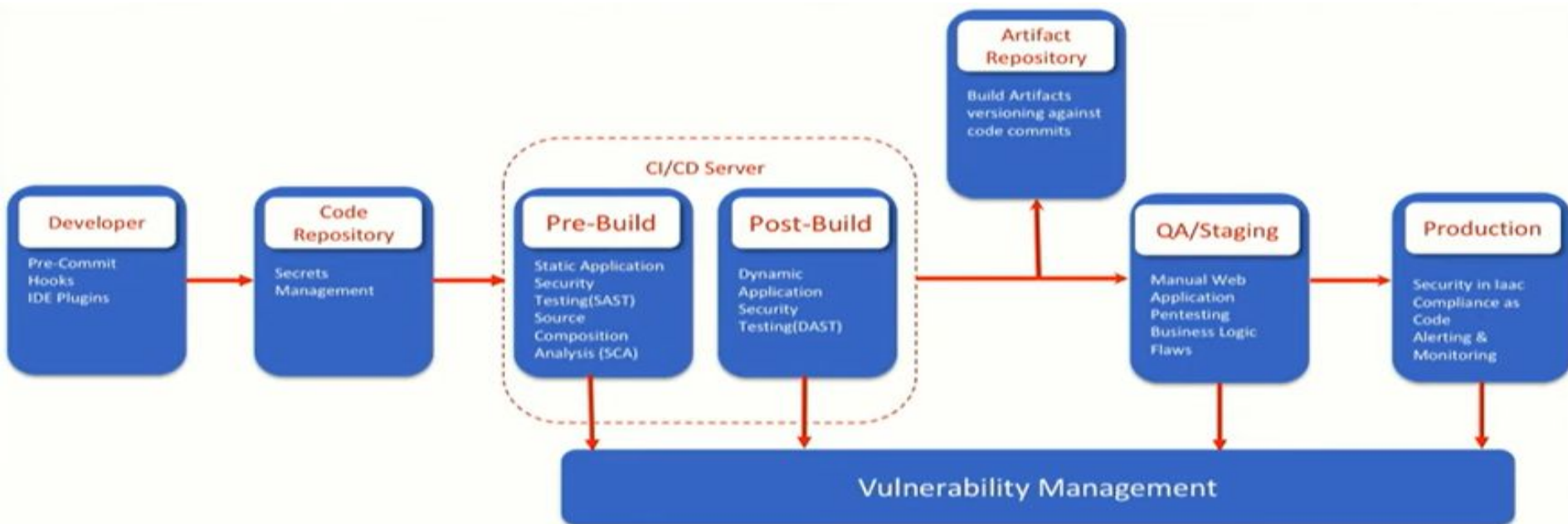


Shifting Left

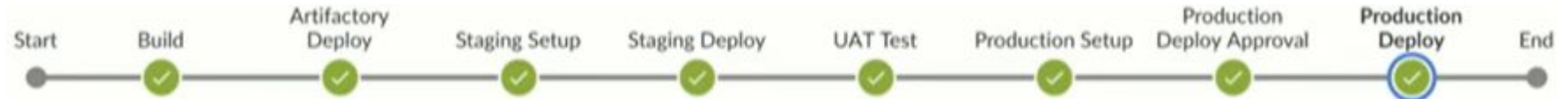


How will you go about it

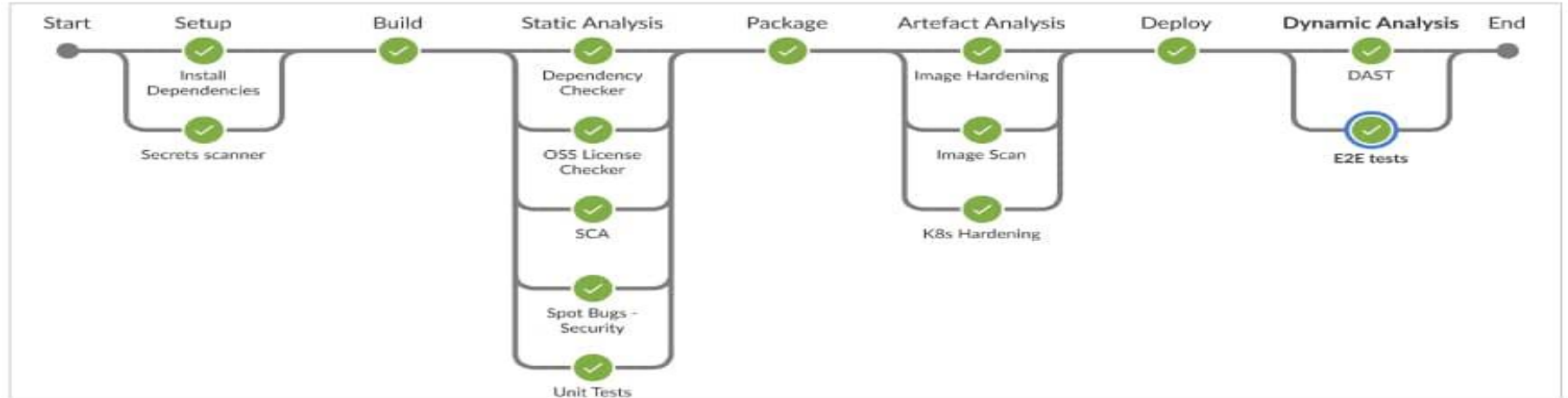
- Automation + Culture Change
- Integrate Security tools into your Devops Pipeline
- Processes



DevOps Pipeline



DevSecOps Pipelines



DevSecOps

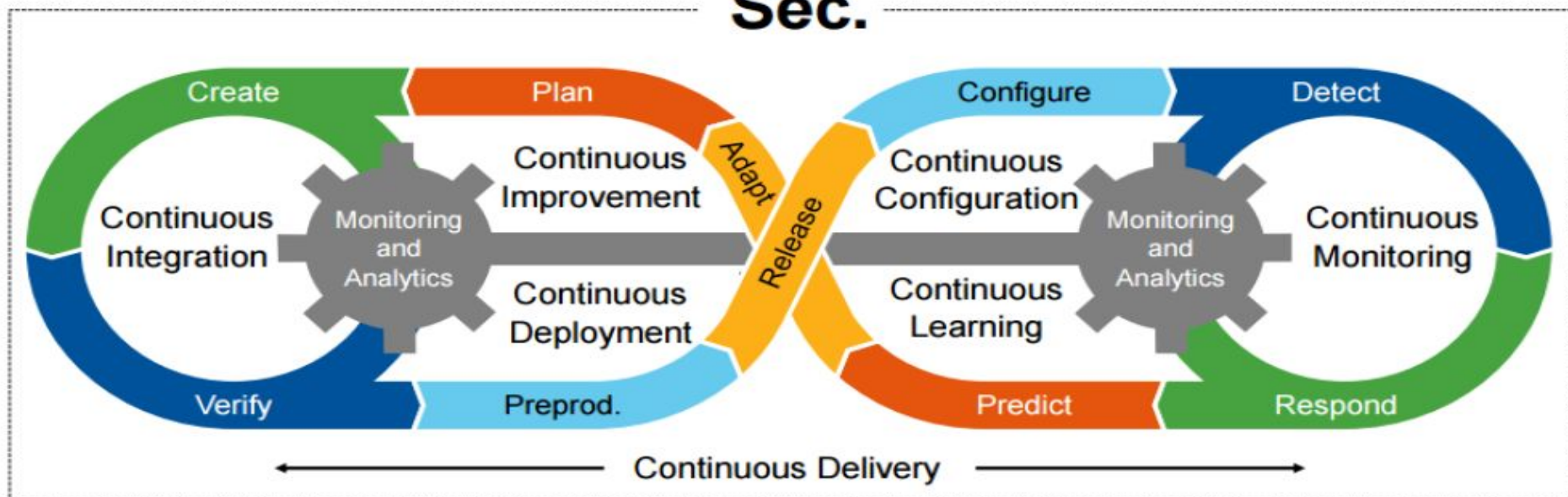


DevSecOps Pipeline

- Pre Commit Hooks
- IDE Security Plugins
- Secrets Management Interface
- Software Composition Analysis
- Static Analysis Security Test (SAST)
- DAST
- Security Infrastructure as Code >>SAST Containerize (Base Images)
- Compliance as Code (PCI DSS) Write Script
- Vulnerability Management (Single source of Truth for Document and Tools)
- Alerting & Monitoring
- Assets Management

Turning DevOps into DevSecOps Seamless Integration with Security

Sec.



it's all about maintaining agility for developers ... without getting in the way

Application Architecture

Layered Architecture

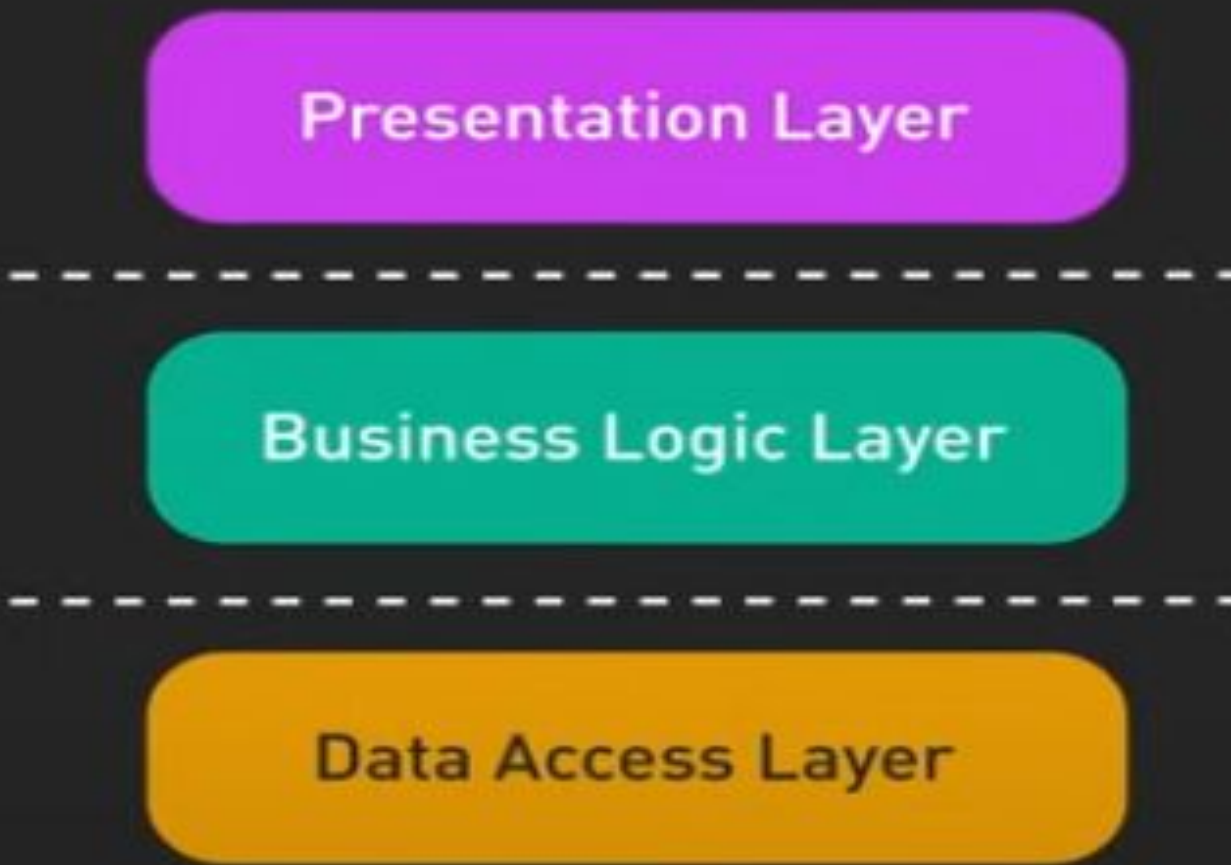
Presentation Layer

Business Layer

Persistence Layer

Database Layer

Presentation Layer

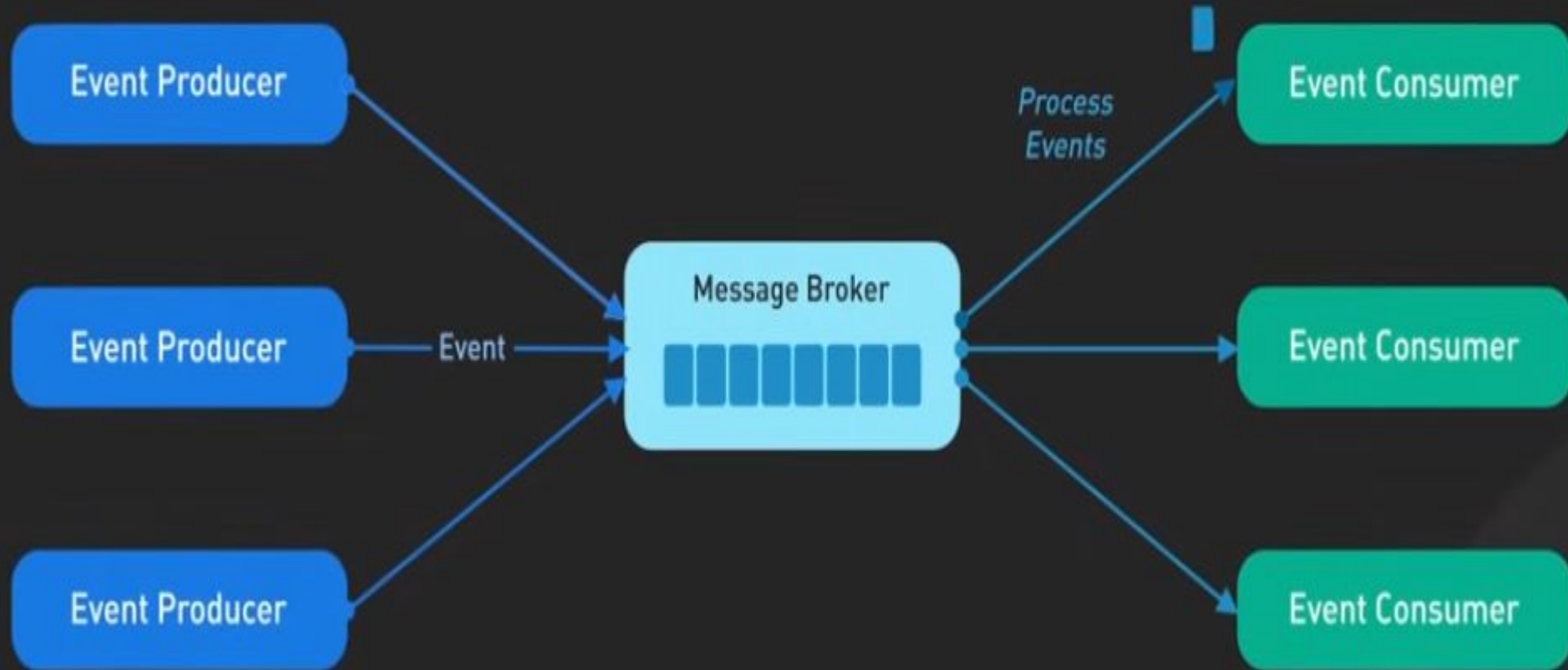


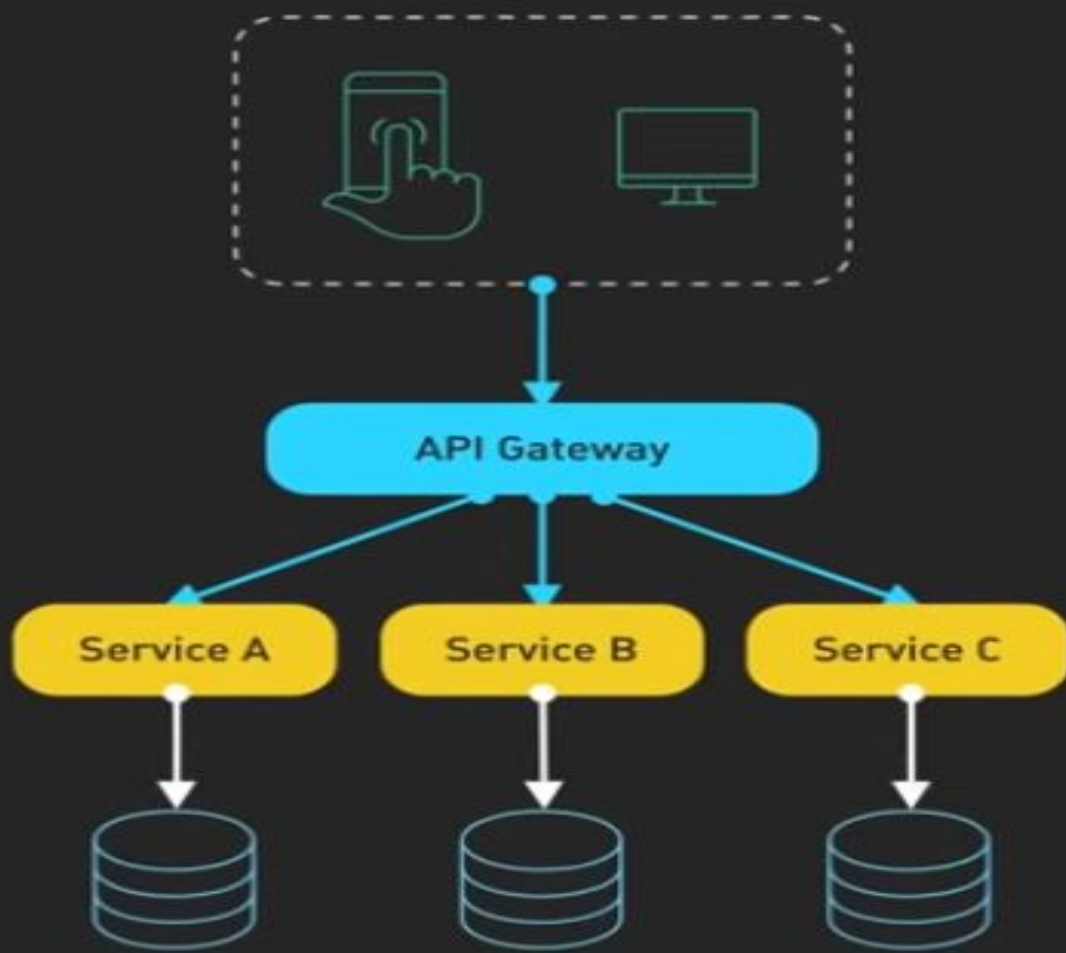
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graph TD; A[Presentation Layer] -.- B[Business Logic Layer]; B -.- C[Data Access Layer];
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Business Logic Layer

Data Access Layer

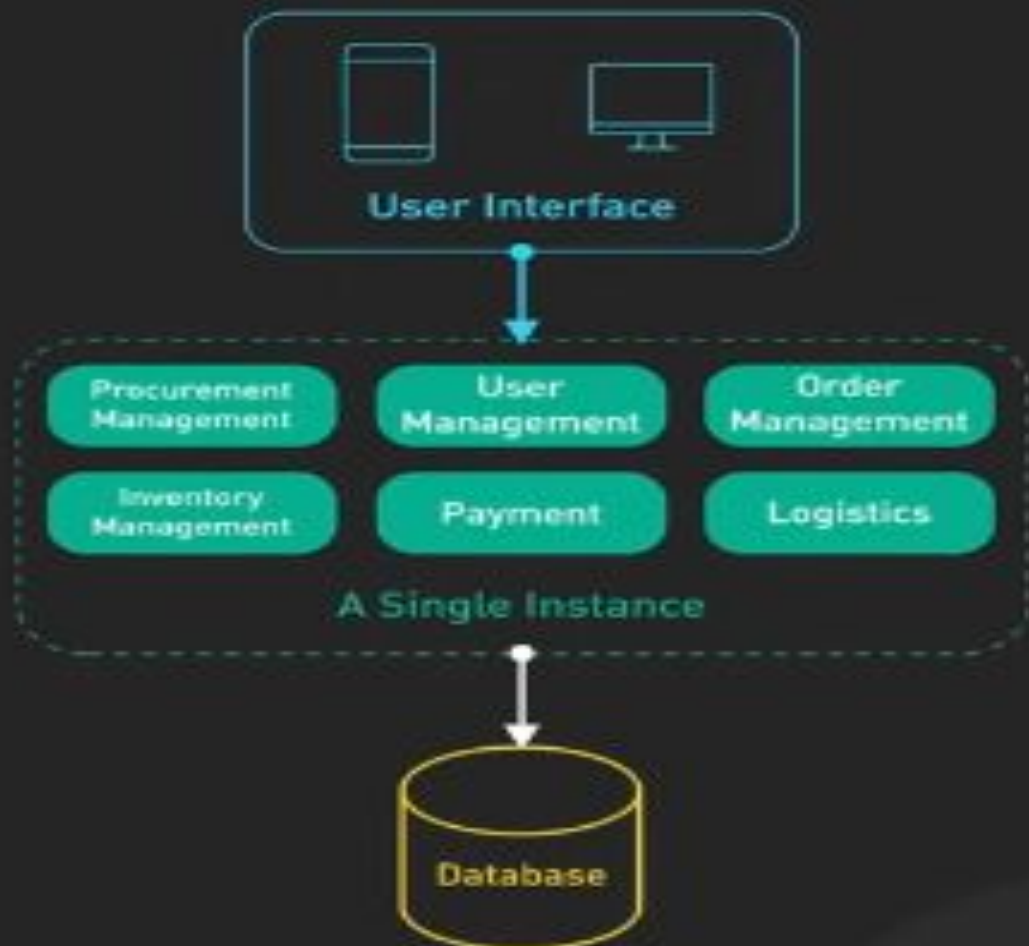
Event Driven Architecture



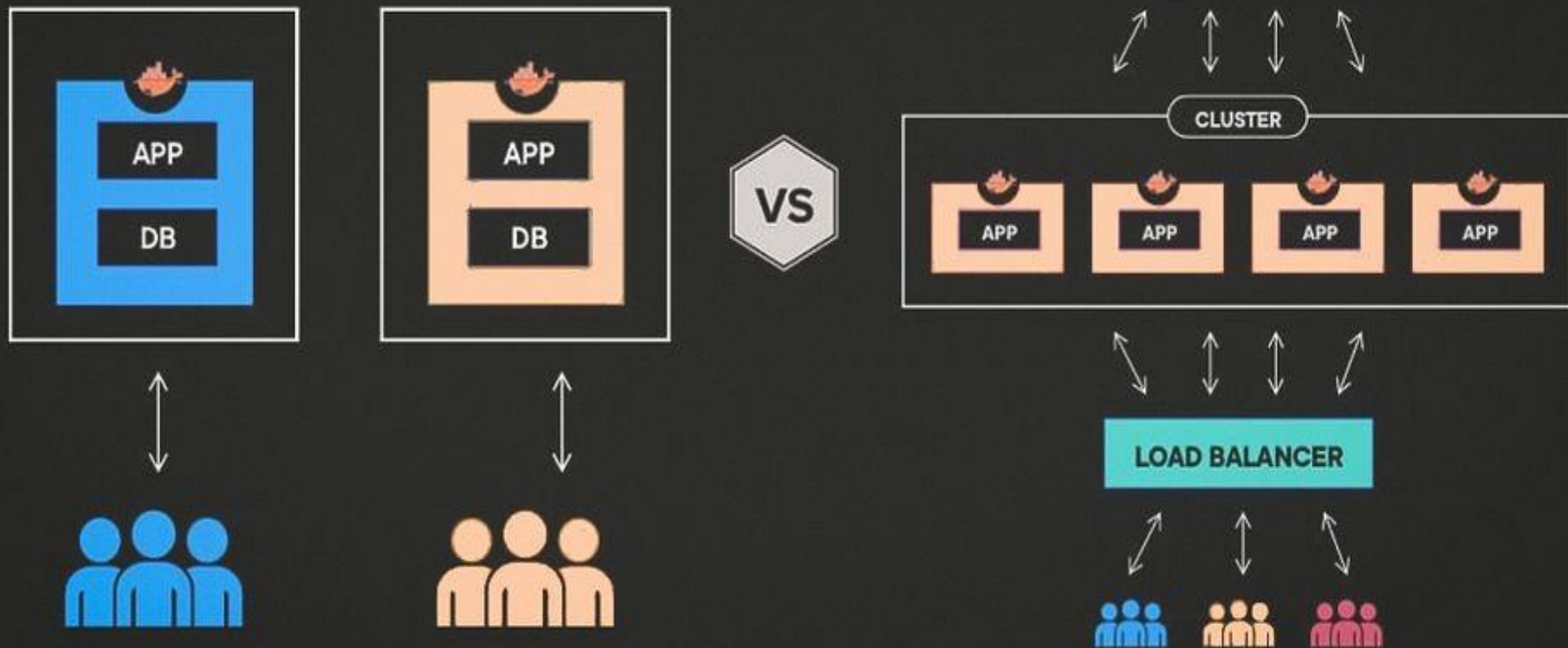


Microservices

Monolithic Architecture



Multi Tenant Architecture



Devops Key Concepts

CI/CD Pipeline Setup- CI to automate the build, test, and CD- deployment stages. Ensures continuous integration and delivery of code to Production

Infrastructure as Code (IaC) -Define infrastructure requirements in code, allowing teams to version control and automate the provisioning of cloud infrastructure.

Docker Containerization -Containerize an application with Docker to create isolated environments for running applications, making them portable and reducing conflicts across environments.

Kubernetes for Container Orchestration -Deploy applications in Kubernetes clusters to automate container deployment, scaling, and management, improving resource utilization and reliability.

Load Balancing and Auto-Scaling- Implement load balancing and auto-scaling in a cloud environment (e.g., AWS or GCP) to distribute traffic and automatically adjust resources based on demand.

Continuous Feedback Loop with DevOps- Establish feedback loops with end users, QA teams, and stakeholders throughout the SDLC to improve product quality and responsiveness.

Monitoring-Integrate ChatOps tools (e.g., Slack, Microsoft Teams) with CI/CD pipelines and monitoring tools to enable real-time collaboration and incident response within a team